

Shaurya Pathak

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Education

University of California, Berkeley

Aug 2025 – Present

Master of Information and Data Science

Focus: Machine Learning, Scalable Data Systems, and Statistical Inference

University of California, Riverside

Sep 2021 – Dec 2024

B.S. Computer Science with Business Applications — GPA: 3.82 (Major: 3.88)

Chancellor's and Dean's List; Undergraduate Research Assistant to Prof. Hamed Mohsenian-Rad

Experience

NVIDIA

Santa Clara, CA

Software Development Engineer in Test

Mar 2025 – Present

- Architected automated debugging orchestration system across hundreds of release workflows using AI-driven analysis, Celery, RabbitMQ, and Python, reducing false positive reports by 40% and freeing engineering capacity for strategic work.
- Built multi-platform validation infrastructure for GPU driver stacks ensuring numerical accuracy through distributed benchmarking of latency, throughput, and floating-point precision under varied workloads.
- Engineered CI/CD pipelines with automated regression testing, reducing validation runtime by 60% while expanding test coverage for cross-architecture compatibility.

Data Engineer – NeMo (Generative AI)

Aug 2024 – Mar 2025

- Designed high-throughput ETL pipelines and analytical dashboards supporting LLM training and evaluation, enabling data-driven model refinement across research teams.
- Developed 15+ statistical metrics quantifying annotation bias, linguistic diversity, and dataset consistency, directly informing model iteration strategies and data quality standards.
- Collaborated with ML researchers to implement automated validation workflows, accelerating model iteration and establishing reproducible evaluation frameworks.

NASA Jet Propulsion Laboratory (City of Los Angeles Project)

Remote

Student Researcher

Sep 2021 – Dec 2024

- Built time-series prediction models using convolutional LSTM to forecast PM_{2.5} from satellite imagery and sensor data, achieving RMSE of 1.6 $\mu\text{g}/\text{m}^3$ without ground stations.
- Engineered automated AWS inference pipeline (Glue, Lambda, S3) generating hourly pollution forecasts with web-based visualization at scale.
- Integrated static and real-time traffic datasets into simulation frameworks for congestion analysis and urban planning studies.

AllCare Therapies

Cerritos, CA

Software Engineer Intern

Jun 2024 – Aug 2024

- Developed serverless ML classification pipeline using AWS Lambda and Step Functions, automating 90% of document processing workflows.
- Optimized data ingestion and model inference throughput, delivering 3× performance improvement through pipeline refinements.

Technical Skills

Languages: Python, SQL, C++ (familiar), Bash

Libraries & Frameworks: Pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, Matplotlib, Celery, RabbitMQ

Systems & Tools: AWS (Lambda, Glue, S3, Step Functions), Linux, Docker, Git, Ansible, CI/CD, CUDA

Domains: Research Platform Infrastructure, Statistical Modeling, Data Pipeline Engineering, Distributed Systems, Simulation & Validation Frameworks